

Intentional Design Choices

Most VAE visualizations are static, just showing a fixed latent space. I intentionally designed **Draw with VAE** to be a fun lab where users can interactively draw and observe how small changes in their drawn digits affect encoded mappings and how sampling distributions and KL divergence greatly impact the latent space. A key choice was the **"Gold Glow"** and **"KL Drift"** indicators. These provide immediate visual feedback on the model's learned probability density, transforming abstract KL Divergence math into a visible "comfort zone" of the model. I also included the **4-Neighbor View** to prove that the VAE learns a continuous neighborhood—a concept of a digit—rather than the isolated "memory vaults" seen in standard Autoencoders.

Effectiveness and Utility

The demo is effective because it directly contrasts the **shattered manifold** of an AE with the **continuous manifold** of a VAE. By allowing users to toggle $\beta=0$ (AE), they see the coordinate "teleport" away from the origin, visually confirming the lack of regularization. Furthermore, the **Distributional Robustness** tests (Laplace, 'The Edge', etc) allow users to see exactly where the model's "imagination" in decoding fails, making it a powerful diagnostic tool for understanding generative limits.

Iteration and Implementation

The project iterated through several technical phases. I first sourced a robust generative model with weights compatible with a JavaScript environment. I then utilized Gemini to refine the UI's aesthetic—specifically for the real-time density rendering—and Claude to streamline the connection between the model weights and the interactive canvas. Early iterations lacked clarity on sampling, so I added **synthetic scatter plot previews** in the hover states to show the different priors before a user commits to sampling from them.